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- An $(\mathbf{x}^*, \mathbf{u})$ (or $\mathbf{x}^*$) which satisfies the **Karush Kuhn Tucker (KKT)** conditions is called a **KKT** point.

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- **Remark:** If $\mathbf{x}^*$ is a **(KKT)** point then it is also an **FJ** point.
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- Take $u_1 = \frac{1}{3}$, $u_2 = \frac{2}{3}$, then
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Hence $[2, 1]^T$ satisfies the **KKT** condition.

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- So **KKT** condition is **not** a necessary condition for a **local minimum**, **FJ** conditions are.

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 - 2 Is $[0, 0]^T$ a local minimum point for this problem? If not find $\mathbf{d} \in D_{\mathbf{x}^*}$ along which the function f will decrease.

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 - Is $[0, 0]^T$ a local minimum point for this problem? If not find $\mathbf{d} \in D_{\mathbf{x}^*}$ along which the function f will decrease.
- Minimize $(x_1 - 2)^2 + (x_2 - 3)^2$
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 $x_1^2 + x_2^2 \leq 5$.
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- Consider the following problem:

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- For a problem **(P)** of the form,

Minimize $f(\mathbf{x})$

subject to $g_i(\mathbf{x}) \leq 0$, for $i = 1, \dots, m$, $\mathbf{x} \in \mathbb{R}^n$,

where all the g_i 's and f are continuously differentiable throughout $\mathbb{R}^n$, and S is the feasible region of **(P)** check the correctness of the following statements with proper justification.

- 1 If $\mathbf{x}^* \in S$ is a KKT point of the above problem then $-\nabla f(\mathbf{x}^*)$ lies in the cone generated by $\nabla g_i(\mathbf{x}^*)$, $i \in I$, where I gives the indices of the binding constraints (given by g_i 's) at $\mathbf{x}^*$.

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- 4 If $\mathbf{x}^*$ is an FJ point and there is a solution to the FJ conditions at $\mathbf{x}^*$ with $u_0 = 0$ (u_0 is the coefficient of $\nabla f(\mathbf{x}^*)$ in the FJ conditions), then $\mathbf{x}^*$ is not a KKT point.

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- If all the g_i 's and f are convex functions and $\mathbf{x}^* \in S$ is such that $F_{0,\mathbf{x}^*} \cap G_{0,\mathbf{x}^*} = \phi$, then $\mathbf{x}^*$ is a global minimum of f in S .

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- If $\mathbf{x}^*$ is a local minima of (P) with $F_{0,\mathbf{x}^*} \neq \phi$ but $G_{0,\mathbf{x}^*} = \phi$ then $\mathbf{x}^*$ is not a KKT point.

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- There exists a (P) with $Fea(P) \neq \phi$ such that $D_{\mathbf{x}^*} = \phi$ for all $\mathbf{x}^* \in S$.
- Give examples of nonconstant functions on $\mathbb{R}^n$ which are both convex and concave and those which are neither convex nor concave.

- For a linear programming problem (P) of the form,
Minimize $\mathbf{c}^T \mathbf{x}$ subject to $A_{m \times n} \mathbf{x} \leq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$,
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Solve the KKT conditions for u_i 's. Hence find an optimal
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- Check whether $[1, 0]^T$ is a KKT point of the above problem. How many feasible points does this problem have? What is your conclusion?